Mohamed Fakhry Eldin Ghalwash, Ph.D.

https://m-fakhry.github.io/ | mohamed.fakhry@gmail.com
Citations | +2 010 5000 3412
ORCID | Scopus and Web of Science



About

Mohamed Ghalwash is a machine learning researcher focused on developing novel, interpretable models for spatio-temporal pattern discovery in complex event data, with applications spanning healthcare analytics, health behavior modeling, and decision support systems that generate real-world evidence from medical records, wearable devices, and instrumented environments.

At IBM, he led a research agenda that introduced novel machine learning techniques into IBM's healthcare and life sciences offerings, which ranged from disease progression modeling to new approaches for chronic disease management. His work on temporal sequence mining and multitask learning gave clinicians and researchers an objective, data-driven understanding of how complex chronic diseases progress. This research has informed drug development strategies aimed at slowing or halting disease progression. He also contributed to machine learning frameworks that incorporate external domain knowledge alongside data.

Prior to IBM, at Temple University, he developed machine learning models for early and accurate detection of sepsis onset and contributed to predictive models for temporal social networks and large-scale temporal graphs using probabilistic exponential graphical models, while mentoring graduate students.

At Proteinea, he led a team of AI researchers, data scientists, and engineers, and built a structured learning-and-development framework that measurably improved team capability and research output.

As an academic supervisor, he has supervised eight postgraduate students (one PhD and seven MSc) across the Faculty of Science and the Faculty of Engineering at Ain Shams University, and Zewail City of Science, Technology, and Innovation. Their thesis topics span EEG signal classification, cryptanalysis and side-channel security, steganography, structure learning in Bayesian networks, and explainable AI for image-based concept localization. He also serves as co-PI on a three-year STDF-funded grant developing AI-driven immunoinformatics approaches for Foot-and-Mouth Disease vaccine design.

His long-term goal is to advance the fields of machine learning and pattern recognition through original research contributions and to translate these ideas into real-world applications and products.

Achievements: Published 55+ peer-reviewed papers (17 journal articles, 31 conference papers, 1 book chapter, 6 conference abstracts; highest IF 46). 8 patents. 6 research awards (2 IBM Patent Plateau Awards, 2 Ain Shams University Publication Awards, 2 Best Graduate Paper Awards). 4 journal editorial boards. Program committee member for NeurIPS, ICML, ICLR, AAAI, and IJCAI.

Employment History

Academia

2025/9 – Present ◆ **Associate Professor,** Data Science and AI Program, School of Computational Sciences and Artificial Intelligence, Zewail City, Giza, Egypt (on secondment from Ain Shams University).

2023/1 – Present ◆ **Associate Professor,** Computer Science, Mathematics Department, Faculty of Science, Ain Shams University, Cairo, Egypt.

Employment History (continued)

2014/9 – 2016/8 **Research Assistant Professor,** Center for Data Analytics and Biomedical Informatics, Temple University, Philadelphia, USA.

2014/6 – 2022/12 **Assistant Professor,** Faculty of Science, Ain Shams University, Cairo, Egypt (on leave, 2014/9 – 2023/7, for appointments at Temple University and IBM).

2013/9 – 2014/8 **Postdoctoral Associate,** Center for Data Analytics and Biomedical Informatics, Temple University, Philadelphia, USA.

2011/9 – 2013/8 **Research Assistant,** Center for Data Analytics and Biomedical Informatics, Temple University, Philadelphia, USA.

1998/10 – 2009/8 **Teaching Assistant,** Faculty of Science, Ain Shams University, Cairo, Egypt.

Industry

2023/8 – 2024/8 **Director of AI and R&D,** Proteinea, Egypt.

2016/10 – 2023/7 **(Senior) Research Scientist,** IBM T.J. Watson Research Center, NY, USA.

2008/2 – 2009/8 **Technical Team Leader,** Asset Technology Group, Cairo, Egypt.

2005/7 – 2008/1 **Java Team Leader, then Java Architect,** Raya Software, Cairo, Egypt.

2002/1 – 2005/6 **Software Developer and Instructor,** Nile Soft, Cairo, Egypt.

2001/1 – 2001/8 **Java Instructor,** Allied Consultant, Cairo, Egypt.

Education

2009/9 – 2013/8 **Ph.D. in Computer Science (Machine Learning and Pattern Recognition)**.
Temple University, Philadelphia, USA.
Thesis title: *Interpretable Early Classification of Multivariate Time Series.*

2001/2 – 2006/6 **M.Sc. in Computer Science (Randomized Algorithms)**.
Faculty of Science, Ain Shams University, Cairo, Egypt.
Thesis title: *Randomized Algorithms and Its Applications.*

1994/9 – 1998/6 **B.Sc. in Pure Mathematics and Computer Science**.
Faculty of Science, Ain Shams University, Cairo, Egypt.
Excellent with Honours, Ranked 1st.

Research Projects

Ain Shams University

- **Integration of Artificial Intelligence and Immunoinformatics Approach for Developing Novel Vaccine Strategies for Foot and Mouth Disease.**
  Foot and Mouth Disease (FMD) threatens global livestock and food security, and existing vaccines struggle with the virus's high genetic variability and limited cross-protection across serotypes. This project applies AI to design a universal chimeric vaccine, a multi-epitope recombinant protein that spans the VP1, VP2, and VP3 viral proteins and aims to trigger cross-protective immunity across FMDV serotypes. The project also develops a model that predicts outbreak risk from case and virus-detection data. Funded by Egypt's Science, Technology and Development Fund (STDF), 2026 – 2029; Ghalwash serves as co-PI (PI: Dr. Mohamed Hassan).

IBM

- **Patient Representation.**
  Develop and test patient and disease representations for clinical models. Jointly learning embeddings across domains and fine-tuning them for domains with limited data presents significant AI challenges that call for new methodological approaches. We build a comprehensive library of representation-learning methods and develop novel methods to address these issues. These representation algorithms are assessed for their sensitivity to data characteristics, including fairness.
- **Computational Models for Type 1 Diabetes (T1D).**
  Develop novel computational disease models that could be used to identify factors that impact the rapid/slow progression of Type 1 Diabetes ("T1D") in infants and young children who are genetically pre-disposed and may be pre-symptomatic clinically. Such methodologies and models may
  - incorporate heterogeneous features coming from multiple sites and assessments covering multiple aspects of T1D.
  - leverage diverse data sets to accommodate noise and uncertainty in study data.
  - provide a comprehensive view of risk factors that impact the onset of T1D in different time horizons.
- **Objective Pain Metric.**
  This work is focused on identifying correlates of individual pain, defining objective pain metrics, and determining changes to the device parameters that provide the user with the greatest impression of relief. A number of data streams will be studied to identify pain correlates from physiological signals, environmental stresses, and individual behavioral responses. These correlates will be used to identify pain metrics, understand the individual patient's pain experience and how the device parameters can be changed to optimize the sense of change in a manner that is adaptive to the varying needs of the patient.

Research Projects (continued)

Temple University

- **Structured Regression in Complex Networks by Fusion of Qualitative Knowledge and Big Data.**
  Integrating context, domain knowledge, and data-driven modeling of complex phenomena based on observations that are highly structured and interdependent is a very challenging task. In this project we address this problem by using the structured predictive modeling framework based on a probabilistic exponential graphical model instead of relying on traditional statistical approaches that assume independent and identically distributed random variables. Current state-of-the-art predictive modeling techniques usually cannot deal with such large and complex networks; thus we provide knowledge-based compression techniques for complexity reduction. We use multiple kinds of domain knowledge and context to capture additional information that might be missing from the observed data. We also directly constrain the model optimization based on domain constraints and embed other problem-specific qualitative knowledge directly into the framework. The main innovation of this project is extending our structured learning models to explore in detail the hypothesis that a unified approach of integrating big data with sources of high-level knowledge (ontologies, domain-based constraints etc.) is beneficial for predictive modeling of complex phenomena.

- **Prospective Analysis of Large and Complex Partially Observed Temporal Social Networks.**
  The analysis of social networks often assumes a time invariant scenario, while in practice actor attributes and links in such networks evolve over time and are inextricably dependent on each other. In addition, the temporal graph is just partially observed, multiple kinds of links exist among actors, various actors have different temporal dynamics and environmental influence can be both positive and negative. This project is closely examining the hypothesis that a unified approach of jointly modeling these and related problems is beneficial for prospective analysis of large-scale partially observed temporal hypergraphs. Novel methods for analyzing large and evolving graphs developed on the project are evaluated on high impact applications related to predictive modeling of information networks, climate and human health.

- **Predictive Modeling of Patient State and Therapy Optimization.**
  This project develops and validates effective predictive modeling technology to achieve the following sepsis treatment related aims on high dimensional and noisy data at a clinically relevant scale

  - **AIM 1:** Personalized sepsis therapy optimization for an individual patient's state improvement.
  - **AIM 2:** Early diagnosis of sepsis and accurate detection of change in the state of sepsis, and
  - **AIM 3:** Gene expression analysis for sepsis biomarkers identification.

Industry Projects

Asset Technology Group

- **Vodafone DCA (Distribution Channels Automation).**
  Vodafone Egypt (VFE) has established agreements with indirect sales business partners (Agents) and 3rd party (Exclusive Dealers, Trade Key Accounts and Dealers) all over Egypt to address all country markets to ensure Vodafone projects and services availability nationwide. Vodafone Egypt sells its products to agents; agents in turn sell Vodafone products to exclusive dealers and dealers who eventually sell to end customers. The current number of agents is 7 with around 50 branches; the number of exclusive dealers is around 150 shops and the number of main normal dealers is around 7000. Vodafone Distribution Department would like to track VFE products' sales movement performed by distribution channels in order to produce accurate incentive calculations and to be able to generate and view analytical reports on dealers' sales patterns in order to study the market behavior versus various parameters (e.g. time, cities, zones, etc.). This project affects VFE, Agents, Trade Key Accounts, dealers and end customers and it is an online web-based application that can be accessed by VFE Sales staff and Agents for stock management, transactions recording and serialized products tracking. The application also enables VFE sales staff to view agents and dealers information and to evaluate their performance in order to manage their incentives. There are as well integrations with other 3rd party systems like Billing System, TIBCO, etc.
- **Vodafone HandHeld.**
  Vodafone sales representatives use a handheld device to make their sales out transactions at the dealer branch. This project is responsible for providing the extranet functionality to the handheld application through HTTP. The handheld application sends an HTTP request to the extranet application to perform a task, and the extranet application returns the results compressed using HTTP compression. Using the handheld application, the sales agent may conduct sales out transactions and check their stock.
- **Vodafone Website.**
  Implementing pilots with Oracle UCM and making certain adjustments to the workflow, security, fragmentation, etc. to meet our demands. The website can be published to an external web server using the publishing tool.

Raya Software

- **Al Watany Bank (AWB).**
  Implementing multithreading technology on the AWB (Al Watany Bank) middleware and applying the needed modifications.
- **Ministry of Civil Aviation (MOCA) Portal.**
  Designing the framework for the Ministry of Civil Aviation (MOCA) portal, one of the company's largest projects.
- **National Societe Generale Bank Middleware (NSGB).**
  The construction of a new middleware for NSGB, an intermediary tier that shields legacy systems from being transparent to retail and customer-serving delivery channels, was part of this project. The main responsibility of the middleware is to receive requests from the frontend, parse the information from these requests, send inquiries to the backend, receive replies back, process the replies and apply some logic, and finally return the results to the requester. Along with managing the project (resource management, mission planning, logistics, status reports, financial control, etc.), participating in the feasibility study to design the project, and participating in the technical design of the required software changes (specifications and programming support), I was also implementing the middleware's core functionality.

Industry Projects (continued)

- **NSGB Admin Web Tool.**
  Designing and implementing with a team of 4 members the NSGB admin web tool. The tool consists of 4 modules.
  - Subscription module is responsible for the subscriptions of a customer to some products or services.
  - Administration module is responsible for maintaining the user profile that uses the system.
  - Reporting module is responsible for generating reports for all channels and these reports could be exported to Excel sheets and PDF.
  - Security module assigns the roles and the privileges for the users.

Nile Soft

- **Arabization of Invoicing System: Duravit Company.**
  As a Technical Consultant, I was in charge of training a team of four members on Java, Jasper Report, and J2EE, and directing the group responsible for developing the Duravit Company's invoicing system and arabization of the inventory items.
- **Automated Web Based Integration System (ERP) of the Credit Guarantee Company (CGC)**
  Working as a Team Leader, leading a team of 4 members responsible for producing UML class diagrams and detailed sequence diagrams for an automated web-based integration system (ERP) of the Credit Guarantee Company (CGC). CGC was mainly established to provide small businesses with loan guarantees. The system is composed of 8 subsystems (Finance/Accounting, Payroll, Fixed Assets, Credit Guarantees, Micro Credit Unit, Access Rights, Deposits, and Investment) and was created in accordance with UML standards. The system's architecture is multi-tiered, built on J2EE standard and the MVC paradigm, and uses Oracle Database Server for the persistent layer with EJB, Struts for the control layer, and JSP for the presentation layer.

Grants

2026 – 2029 — **Integration of Artificial Intelligence and Immunoinformatics Approach for Developing Novel Vaccine Strategies for Foot and Mouth Disease.**
Science, Technology and Development Fund (STDF), Egypt. 2,700,000 LE.
PI: Prof. Mohamed Hassan; Co-PI: Prof. Mohamed Ghalwash.

Invited Talks

2022/11 — **Optimal Ages to Screen**, The 5th Childhood Diabetes Prevention Symposium - General Population Screening for T1D, Barbara Davis Center for Diabetes - School of Medicine - University of Colorado, CO, USA.

2020/11 — **T1D1 Screening Toolbox**, The 3rd Childhood Diabetes Prevention Symposium - General Population Screening for T1D Current Issues in time of COVID-19, Barbara Davis Center for Diabetes - School of Medicine - University of Colorado, CO, USA.

2012/10 — **Characterization of Intrinsic Disorder in a Large Database of Protein Sequences**, 1st Annual Computational Research on Owlsnest (CROO) Symposium, Temple University, Philadlephia, USA.

2011/7 — **Towards Analysis of Uncertainty in Protein Disorder Prediction**, Institute of Computational Molecular Science, Temple University, Philadlephia, USA.

Synergetic Activities to the Research Community

Editorial Board Member

- 2020 – 2024 Big Data.
- 2016 – Present SM Bioinformatics, Genomics and Proteomics.
- 2015 – Present International Journal of Data Mining and Bioinformatics.
- 2016 – 2019 Journal of Biomedical Engineering and Informatics.

Session Chair

- 2013 International Conference of Data Mining (ICDM).

PC Member

- 2026 NeurIPs.
- 2025 NeurIPs.
- 2024 ICLR, NeurIPs.
- 2023 AMIA, ICML, IEEE Big Data, NeurIPs.
- 2022 AMIA, ICML, IEEE Big Data, NeurIPs.
- 2021 AMIA, IEEE Big Data, International Journal of Medical Informatics.
- 2020 AAAI, AMIA, IEEE Big Data, ICLR, IJCAI, SDM, International Journal of Medical Informatics.
- 2019 AMIA, IEEE Big Data, ICLR, IJCAI, SDM, NeurIPs.
- 2018 AMIA, IEEE Big Data, ICLR, IJCAI, SDM, NeurIPs.
- 2017 IEEE Big Data, SDM, NeurIPs.
- 2015 International Conference on Decision Support System Technology, Conference of Artificial Intelligence in Medicine (AIME), 1st Workshop on Matrix Computations for Biomedical Informatics, International Conference on BioInformatics and BioEngineering (BIBE).

Reviewer

- BioMed Research International Journal
- Machine Learning Journal (special Issue for Health and Medicine)

Ad-hoc Reviewer

- 2015 KDD, ICDM, BigData
- 2014 SDM, KDD
- 2013 SDM, KDD, BigData
- 2012 SDM, ICDM

Awards

- 2023 **Publication Awards**, Ain Shams University, Cairo, Egypt.
- 2022 **Publication Awards**, Ain Shams University, Cairo, Egypt.
- 2021 **Patent Plateau Award**, IBM Research, NY, USA.
- 2019 **Patent Plateau Award**, IBM Research, NY, USA.
- 2013 **Best Graduate Paper**, ACM Future of Computing - Temple University, Philadelphia, USA.
- 2012 **Student Travel Award**, IEEE International Conference on Bioinformatics and Biomedicine.

Awards (continued)

- **Best Graduate Paper**, ACM Future of Computing - Temple University, Philadelphia, USA.

2009 ♦ **Graduate Scholarship**, Ministry of Higher Education, Egypt.

Graduate Students

2022/6 – Present ♦ **Mohamed Saleh,** Co-supervisor with Dr. Wael Zakaria.
M.Sc. thesis title: *Feature Extraction for Electroencephalogram Signals Classification.* Ain Shams University, Cairo, Egypt.

2023/1 – Present ♦ **Mostafa Hanafy,** Co-supervisor with Prof. Hatem Bahig.
M.Sc. thesis title: *Cryptanalysis using Machine Learning.* Ain Shams University, Cairo, Egypt.

2025/2 – Present ♦ **Tarek Ahmed,** Co-supervisor with Prof. Nahed Mokhlis.
M.Sc. thesis title: *Development of a New Circular-Linear Distribution for Structure Learning in Bayesian Networks.* Ain Shams University, Cairo, Egypt.

♦ **Taha Elsayed,** Co-supervisor with Associate Prof. Dieaa Nasr, and Prof. Sohier Khamis.
Ph.D. thesis title: *Intelligent Learning Models for Steganography.* Ain Shams University, Cairo, Egypt.

2026/2 – Present ♦ **Israa Elsayed,**
M.Sc. thesis title: *Distributed Concept Localization in Images.* Ain Shams University, Cairo, Egypt.

2027/1 – Present ♦ **Nesreen Ahmed,**
M.Sc. thesis title: *Machine Learning for Side-Channel Attacks.* Ain Shams University, Cairo, Egypt.

♦ **Aya Nageh,**
M.Sc. thesis title: *KAN for Dark Matter.* Zewail City, Giza, Egypt.

♦ **Mohamed Essam,**
M.Sc. thesis title: *Quantum Machine Learning.* Ain Shams University, Cairo, Egypt.

Teaching

2026 – 2027 ♦ DSAI 402: Reinforcement Learning
DSAI 456: Speech Recognition
DSAI 406: MLOps
CSAI 101: Logic Design

2025 – 2026 ♦ DSAI 402: Reinforcement Learning
DSAI 456: Speech Recognition
DSAI 406: MLOps
CSAI 101: Logic Design

2024 – 2025 ♦ COMP 304: Compiler Design
COMP 402: Bioinformatics
COMP 403: Parallel Computing
COMP 603: Mathematics for Machine Learning

Teaching (continued)

2023 – 2024	♦	COMP 104: Programming Language (C++) COMP 402: Bioinformatics COMP 403: Parallel Computing COMP 603: Mathematics for Machine Learning CS 702: Advanced Artificial Intelligence
2022 – 2023	♦	COMP 104: Programming Language (C++) COMP 303: Formal Syntax and Semantics COMP 402: Bioinformatics COMP 603: Mathematics for Machine Learning COMP 612: Deep Learning
2021 – 2022	♦	COMP 104: Programming Language (C++) COMP 303: Formal Syntax and Semantics COMP 304: Compiler Design COMP 402: Bioinformatics

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